

PARK RAPIDS, MN, USA

WMO: 727453

Lat: 46.901N

Lon: 95.068W

Elev: 437

StdP: 96.18

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.9	(c) -27.7	(d) -33.2	(e) 0.2	(f) -30.0	(g) -30.8	(h) 0.2	(i) -27.1	(j) 12.0	(k) -9.6	(l) 10.9	(m) -9.8	(n) 2.2	(o) 300	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	30.8	21.6	29.0	20.5	27.6	19.6	23.3	28.5	22.1	26.8	21.0	25.8	6.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.3	26.5	20.5	16.0	25.1	19.2	14.7	23.9	71.6	28.4	66.9	26.7	62.9	25.9	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.4	(c) 8.5	(e) -35.2	(f) 33.6	(g) 4.3	(h) 1.5	(i) -38.3	(j) 34.7	(k) -40.8	(l) 35.5	(m) -43.2	(n) 36.4	(o) -46.3	(p) 37.4		
(a) 11.0	(b) 9.4	(c) 8.5	(e) -34.2	(f) 25.3	(g) 3.2	(h) 1.1	(i) -36.5	(j) 26.1	(k) -38.3	(l) 26.7	(m) -40.1	(n) 27.3	(o) -42.4	(p) 28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-13.3	-10.7	-3.4	5.1	12.2	17.6	20.6	19.2	14.5	6.6	-2.0	-9.8
	DBStd	13.17	8.17	7.53	7.44	5.73	4.84	3.71	3.31	3.30	4.83	5.07	6.44	7.31
	HDD10.0	3038	722	579	417	164	32	1	0	0	14	131	362	615
	HDD18.3	5143	981	813	673	398	197	54	17	28	133	364	611	873
	CDD10.0	1137	0	0	2	17	101	231	327	286	148	25	2	0
	CDD18.3	202	0	0	0	1	8	34	86	56	17	1	0	0
	CDH23.3	1982	0	0	0	8	114	347	810	520	168	14	0	0
	CDH26.7	516	0	0	0	1	28	76	236	133	40	2	0	0
Wind	WSAvg	4.0	3.9	3.9	4.1	4.3	4.3	3.9	3.5	3.5	3.9	4.3	4.1	3.9
Precipitation	PrecAvg	642	17	12	29	55	69	106	94	92	73	60	26	18
	PrecMax	1029	50	41	73	149	171	228	294	242	146	166	97	48
	PrecMin	318	1	0	3	5	5	18	9	8	10	5	3	4
	PrecStd	115	11	8	15	30	40	50	57	58	38	46	18	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	7.7	17.8	24.0	29.9	31.6	33.4	32.5	30.5	25.5	17.2	6.4
		MCWB	2.1	4.4	11.7	13.7	18.4	22.5	23.2	22.5	20.5	15.5	11.2	3.7
	2%	DB	1.6	4.4	13.2	20.6	26.4	28.8	30.9	29.7	27.5	21.0	12.9	3.0
		MCWB	0.1	1.9	8.1	10.9	16.4	20.3	22.3	20.9	19.7	14.2	8.4	1.2
	5%	DB	-0.2	2.3	9.3	17.9	23.7	27.0	29.1	27.9	25.1	17.5	9.3	1.2
		MCWB	-1.2	0.6	5.5	9.4	14.7	18.4	21.1	19.8	18.0	12.2	5.9	0.0
	10%	DB	-2.1	0.3	6.1	15.1	21.3	25.3	27.7	26.4	22.7	14.8	6.7	-0.3
		MCWB	-3.1	-0.9	3.2	7.9	13.4	17.6	20.3	18.8	17.0	10.2	4.0	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	4.7	13.3	14.9	20.5	24.1	25.2	24.4	22.6	17.4	12.4	4.1
		MCDB	4.5	7.4	16.0	23.0	27.1	29.7	30.0	30.1	27.9	20.7	15.0	6.5
	2%	WB	0.3	2.4	8.4	12.1	18.1	21.8	23.8	22.4	20.9	15.0	8.7	1.4
		MCDB	1.4	4.0	12.6	17.9	23.5	26.7	29.0	27.3	25.9	20.0	11.9	2.8
	5%	WB	-1.2	0.7	5.7	10.2	16.3	20.3	22.6	21.3	19.4	12.6	6.4	0.2
		MCDB	-0.2	2.1	9.2	16.7	21.6	24.9	27.5	25.9	23.2	16.8	9.4	1.2
	10%	WB	-3.0	-1.0	3.4	8.5	14.6	18.9	21.4	20.2	17.9	10.7	4.1	-1.3
		MCDB	-2.2	0.3	6.1	14.4	19.5	23.3	26.0	24.5	21.7	13.9	6.5	-0.4
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.4	10.9	11.8	12.2	11.3	11.8	11.9	11.6	10.0	8.8	9.0
		MCDBR	9.8	11.0	13.4	16.8	16.2	13.9	13.2	13.8	14.3	14.9	12.9	9.7
		MCWBR	8.5	8.6	8.8	9.2	8.3	6.7	6.3	6.5	7.3	8.7	8.4	8.1
	5% WB	MCDBR	9.6	9.9	12.8	14.7	13.2	11.3	12.0	11.8	11.5	13.2	12.1	9.0
		MCWBR	8.4	8.0	8.6	8.8	7.5	6.7	6.7	6.3	6.8	8.3	8.4	7.7
Clear-Sky Solar Irradiance	taub	0.266	0.305	0.302	0.354	0.370	0.375	0.378	0.380	0.351	0.307	0.280	0.261	
	taud	2.337	2.214	2.390	2.339	2.356	2.367	2.379	2.388	2.451	2.557	2.518	2.400	
	Ebn at Noon	832	855	926	901	895	889	880	864	862	860	822	800	
	Edh at Noon	72	102	100	117	119	118	116	110	94	73	61	61	
All-Sky Solar Radiation	RadAvg	1.59	2.72	3.84	4.84	5.36	5.85	6.23	5.24	3.89	2.39	1.46	1.11	
	RadStd	0.17	0.27	0.39	0.41	0.37	0.40	0.28	0.43	0.29	0.37	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-1.33	-1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page